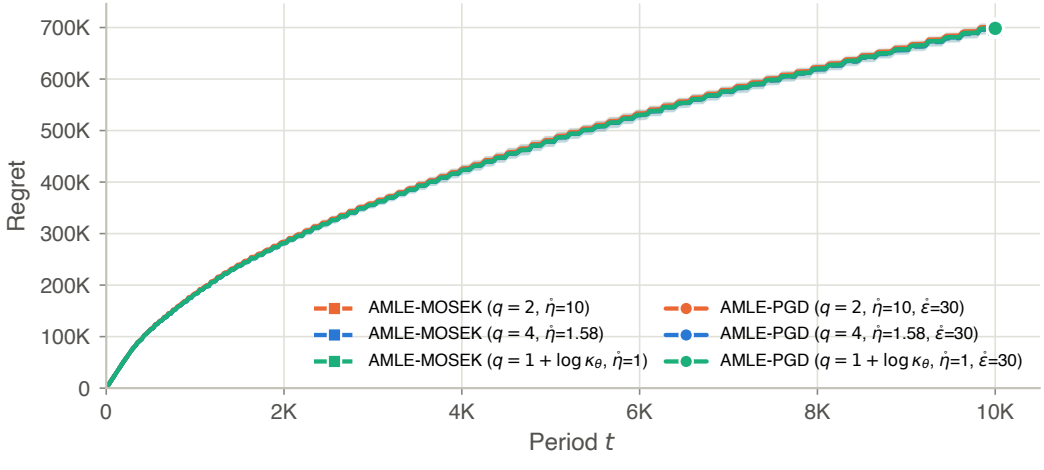
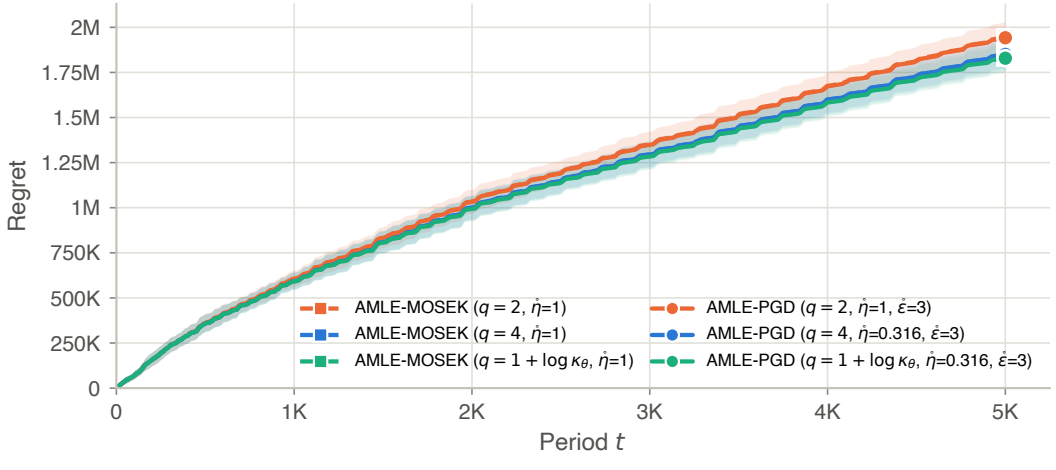

(a) Case I: Section 5.1 design (dense parameters)
Regret
Period t
0
100K
200K
300K
400K
500K
600K
700K
0
2K
4K
6K
8K
10K
AMLE-MOSEK ($q = 2$, $\mathring{\eta}$=10)
AMLE-MOSEK ($q = 4$, $\mathring{\eta}$=1.58)
AMLE-MOSEK ($q = 1 + \log \kappa_\theta$, $\mathring{\eta}$=1)
AMLE-PGD ($q = 2$, $\mathring{\eta}$=10, $\mathring{\varepsilon}$=30)
AMLE-PGD ($q = 4$, $\mathring{\eta}$=1.58, $\mathring{\varepsilon}$=30)
AMLE-PGD ($q = 1 + \log \kappa_\theta$, $\mathring{\eta}$=1, $\mathring{\varepsilon}$=30)


(b) Case II: heavy-tailed features and sparse parameters
Regret
Period t
0
250K
500K
750K
1M
1.25M
1.5M
1.75M
2M
0
1K
2K
3K
4K
5K
AMLE-MOSEK ($q = 2$, $\mathring{\eta}$=1)
AMLE-MOSEK ($q = 4$, $\mathring{\eta}$=1)
AMLE-MOSEK ($q = 1 + \log \kappa_\theta$, $\mathring{\eta}$=1)
AMLE-PGD ($q = 2$, $\mathring{\eta}$=1, $\mathring{\varepsilon}$=3)
AMLE-PGD ($q = 4$, $\mathring{\eta}$=0.316, $\mathring{\varepsilon}$=3)
AMLE-PGD ($q = 1 + \log \kappa_\theta$, $\mathring{\eta}$=0.316, $\mathring{\varepsilon}$=3)